

Project Breakdown: Emotiv EPOC X EEG Pipeline

The Big Picture

The client wants a completely offline MATLAB program that takes raw brainwave data from a specific headset (the Emotiv EPOC X), automatically cleans up the noise, and generates visual graphs of that data. The most important rule is that everything must happen directly on the computer—no cloud storage or internet processing is allowed.

What You Need to Build

- **An automatic cleaner:** A script that takes raw brainwave files (EDF or BDF format) and automatically filters out noise, like electrical interference, bad signals, or eye blinks.
- **A local save system:** The program must save the raw and cleaned data in organized folders directly on the user's hard drive.
- **A graph viewer:** A tool that calculates the "power spectrum" (how strong the brainwaves are between 0-20 Hz) and displays it visually for all 14 channels of the headset.
- **A comparison tool:** The viewer must be able to display three specific recordings side-by-side to track changes over time: the very first recording, the second-to-last recording, and the final recording.

What You Need to Deliver

When the project is finished, you must hand over:

- **The Code:** All the MATLAB/EEGLAB scripts required to run the automatic cleaning and generate the graphs.
- **Code Notes:** Clear comments inside the code explaining to the client how it works.
- **Batch Processing:** A feature that allows the script to handle multiple recordings at the same time.
- **A README File:** A brief instruction manual giving the client clear directions on how to run your program.

What You Must Include in Your Bid

When you apply for this job, the client wants you to explicitly state these five things in your proposal:

1. Confirm that your final program will run 100% locally and offline.

2. Confirm that your script can automatically grab the data file, process it in the background, and pop up the final graph.
3. Confirm that you have previous experience cleaning EEG (brainwave) data.
4. Confirm that you can make your final graphs look exactly like the reference picture they provided.
5. Provide an estimate of how long it will take you to finish the job.